Damped two-stream and twin couplings in equilibrium reasoners: attractor geometry, inner loops and test-time scaling on Sudoku-Extreme

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Abstract

Equilibrium Reasoners (EqR) iterate one shared block on two latent states and reach high accuracy on Sudoku-Extreme by spending test-time compute: more iterations (depth), more random starts (breadth), and Langevin noise. We replace EqR's self-reading update by two-stream *kick* and *relax* couplings in which each stream is driven only by the other, so that every update is an invertible shear followed by a constant scaling. We add a *twin* coupling, in which the low-level state is doubled into two copies that read each other; it restores EqR's inner fixed-point iteration while every update still reads only streams it does not change. We study which properties of the resulting maps decide accuracy, depth scaling, breadth and the effect of noise. On the theory side we show that (i) the force block has unit per-token RMS, so the state at a fixed point is an exact, rate-determined multiple of a unit vector, which fixes the scale against which test noise acts; (ii) the volume constraint of the coupled maps is not binding for any trained model except one with near-identity rates; (iii) symmetric relaxation without common-mode damping has an exactly neutral mode, and damping it is what makes relax work; (iv) alternating couplings have no implicit inner fixed-point problem, while EqR's inner loop does; (v) a relaxed self-map with strong negative feedback undergoes a period-doubling (flip) bifurcation; and (vi) the twin's inner loop is a lag-two recurrence on one interleaved sequence, whose fixed points are the fixed points *and* 2-cycles of the self-reading map, so a flip of the self-map becomes a symmetry-breaking pitchfork of the twins. Numerically, on 14 checkpoints we measure Jacobian spectra, path independence, selection, and escape under noise. Right answers are attracting fixed points and wrong end states are not, in every working model, so convergence-based selection is nearly perfect and accuracy is limited by *coverage*. Test noise helps by selectively destabilising wrong attractors below an arm-specific breakdown level. EqR differs from the alternating couplings in being near critical at its answers ($\rho \approx 0.93$–0.97 per outer iteration against 0.53–0.73), in being more path independent, and in an inner loop that, with training, goes through a flip bifurcation into an exact 2-cycle. After 20k training steps the twin coupling with damping $\delta = 2$ is the best test-time scaler we measured: 98.2% exact at depth 1024 and 99.6% top-4 with 128 starts at depth 64, above EqR's released 50k-step checkpoint (97.6%, 98.6%), although it is the weakest working arm at 16 iterations and its gain comes from test noise. Trained to 50k steps it generalises like EqR (training and test accuracy within one point, against gaps of 15–26 points for the alternating couplings, which fit the 1000 training puzzles), but loses part of its noise-driven depth scaling, and EqR leads at 50k. All results are single-seed.

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1 Introduction

Looped (weight-tied, recurrent-depth) reasoners solve constraint problems such as Sudoku by iterating one block many times [8, 13, 14, 19]. Equilibrium Reasoners (EqR, [13]) add three ingredients to TRM-style recursion: random initial states, Langevin noise in the state, and an explicit view of reasoning as convergence to an attractor. At test time EqR scales *depth* (iterations), *breadth* (random starts per puzzle, with an answer chosen by convergence) and noise.

This note asks which properties of the iteration map make these work. We keep EqR's data, block, training recipe and evaluator, and replace only the state update by two-stream couplings in the spirit of conformal symplectic integrators [7, 16] and momentum networks [18]:

$$\begin{aligned} \textbf{kick:}\quad & L \leftarrow a\,(L + \mathcal{B}(H + x)), && H \leftarrow bH + \mathcal{B}(L + x), \\ \textbf{relax:}\quad & L \leftarrow aL + (1-a)(cH + \mathcal{B}(H + x)), && H \leftarrow bH + (1-b)(cL + \mathcal{B}(L + x)), \end{aligned}$$

with $a = e^{-\gamma}$, $b = e^{-\alpha}$, $c = e^{-\delta}$. Each stream is driven only by the other stream, so every update is invertible with a state-independent Jacobian determinant. The price is that L, given H, is one explicit layer: EqR's inner iteration disappears (Prop. 6). The *twin* coupling (§2.3) brings it back without giving up invertibility by doubling L into two copies that read each other,

L reading L' where EqR's z_L reads itself:

$$\textbf{twin:}\quad L \leftarrow aL + (1-a)\big(cL' + \mathcal{B}(L' + H + x)\big),\quad L' \leftarrow aL' + (1-a)\big(cL + \mathcal{B}(L + H + x)\big),$$
$$H \leftarrow bH + (1-b)\big(cm + \mathcal{B}(m)\big),$$

with $m = \frac{1}{2}(L + L')$. We compare these maps with EqR in accuracy (§4), test-time scaling (§5) and a direct dynamical analysis of trained checkpoints (§6), guided by a small amount of theory (§3).

Summary of findings.

1. **Certification is easy; coverage is hard.** In every working model right answers are attracting fixed points ($\rho < 1$) and wrong end states are not ($\rho > 1$). Convergence, the halting head and a Sudoku validity check therefore pick the right start almost whenever one exists (AUROC ≈ 1); what separates models is how often random starts land in the right basin (§6.1).
2. **Noise acts on wrong attractors first.** Under noise σ below an arm-specific breakdown level $\sigma^\star$ right answers are never lost, while 15–33% of wrong end states are rescued (§6.3), as in Hopfield networks at finite temperature [1, 6]. $\sigma^\star$ depends on the state scale, which is fixed exactly by the rates (Prop. 2), and on how strongly the answers attract.
3. **EqR is near critical and more path independent.** EqR's right answers have $\rho \approx 0.87 \to 0.93 \to 0.97$ (20k, 50k, released) per outer iteration, matching the 0.89–0.97 reported for converged Sudoku loopers [15]; the coupled maps contract much harder (0.53–0.73). EqR's random starts reach fewer distinct answers (1.6 per puzzle at 50k against 2.6 for relax and 3.2 for kick).
4. **EqR's inner loop flips.** With z_H frozen, EqR's inner map has a stable fixed point at 20k steps but sits on an exact 2-cycle at 50k (Prop. 8); its leading Jacobian eigenvalue moves from $\approx \pm 0.6$ to ≈ -1 and, in the released checkpoint, to ≈ -2. The cycle is invisible to the readout: odd inner-loop lengths work as well as even ones.
5. **Damping matters, volume does not.** The volume bound of the coupled maps (Prop. 4) is irrelevant for all trained models except a near-identity kick; relax without common-mode damping has an exactly neutral mode (Prop. 5) and fails, and early-training accuracy improves as the merged-state retention falls from 1 to ≈ 0.2 (§4).
6. **Training length favours EqR over the alternating couplings.** At 20k steps relax ties EqR with enough test-time compute; at 50k EqR leads on every metric (depth 1024: 97.1 vs. 92.5; depth 64 with 128 starts: 98.7 vs. 96.6).
7. **Explicit inner equilibria fit the training puzzles.** At 50k the alternating couplings reach 93–97% on training batches but 71–79% on the test set, and relax $\delta = 2$ declines after 35k steps. EqR, the twin and the self-reading inner loop, the three arms with an implicit inner problem, have no such gap (§4).
8. **A twin inner loop gives the best test-time scaling at 20k.** The twin with $\delta = 2$ is the weakest working arm at 16 iterations (52.3% at $\sigma = 0.5$) but its error falls 26-fold by depth 1024 (98.2%), and with 128 starts at depth 64 it reaches 99.6% top-4 and 99.8% by vote, above EqR's released 50k checkpoint (98.6, 98.7). The gain comes from test noise: at the training noise it reaches only 65.2% at depth 256. At 50k it is accurate without noise (84.6% at 16 iterations, EqR 85.8) and $\sigma = 0.5$ costs it three points (§5.1).

Table 1: Arms. Rates are fixed unless stated. "Positive" kick learns $\gamma = 0.05 + \text{softplus}(\cdot)$ from $\gamma = 1$. Invertible: every update is a shear followed by a constant scaling.

arm	update	rates	invertible	round
EqR	self-read, order L^6H	$1 - \lambda = 0.05$	no	ref.
kick: fixed, learned, positive	kick	γ=α=0.005; learned; learned ≥ 0.05	yes	1
relax, $\delta \in \{0, \frac{1}{4}, 1\}$	relax	$\gamma = \alpha = 3$	yes	1
relax / kick, $x \to L$ only	no x in H's force	as above	yes	2
relax δ=1, $x \to L$, self-read	forces read own stream	$\gamma = \alpha = 3$	no	2
relax, $\delta \in \{2, 3\}$	relax	$\gamma = \alpha = 3$	yes	3
twin L, $\delta \in \{1, 2\}$	L, L' read each other, order $LL'LL'LL'H$, $x \to L$ only	$\gamma = \alpha = 3$	yes*	3
inner loop, self-read	self-read relax, order L^6H	$\gamma = \alpha = 3$, $\delta = 1$	no	3

*Every update reads only streams it does not change, so the inverse is explicit (Prop. 9); it is not implemented in code.

2 Setting

2.1 EqR

A puzzle is embedded as $x \in \mathbb{R}^{n \times d}$ ($n = 97$ positions: 16 puzzle-embedding slots and 81 cells; $d = 512$). The shared block $\mathcal{B} : \mathbb{R}^{n \times d} \to \mathbb{R}^{n \times d}$ is two post-norm layers, each $v \mapsto N(v + \text{MLP}_t(v))$ followed by $v \mapsto N(v + \text{SwiGLU}(v))$, where MLP_t mixes tokens and N is RMS normalisation *without gain*, $N(y)_i = y_i / \text{rms}(y_i)$ row-wise.

Definition 1 (EqR update [13])**.** With $\lambda = 0.95$ and noise $\xi \sim \mathcal{N}(0, \sigma^2 I)$ drawn afresh after every update, one *latent recursion* is

$$z_L \leftarrow (1 - \lambda) z_L + \lambda\, \mathcal{B}(z_L + z_H + x) + \xi \quad \text{(6 times)}, \qquad z_H \leftarrow (1 - \lambda) z_H + \lambda\, \mathcal{B}(z_H + z_L) + \xi.$$

One *outer iteration* (one ACT step) is three recursions, i.e. 21 block evaluations; the readout decodes z_H. Training differentiates only the last recursion (7 evaluations), carries the state across outer iterations detached, starts from random states (truncated normal, std 1), and uses $\sigma = 0.01$.

2.2 Coupled updates

Definition 2 (Kick and relax)**.** Let $a = e^{-\gamma},\ b = e^{-\alpha},\ c = e^{-\delta} \in (0, 1]$. A *half-step* updates one stream from the other:

$$\begin{aligned} \text{kick:} \quad & L \leftarrow a\,(L + \mathcal{B}(H + x)), && H \leftarrow bH + \mathcal{B}(L + x), \\ \text{relax:} \quad & L \leftarrow aL + (1 - a)(cH + \mathcal{B}(H + x)), && H \leftarrow bH + (1 - b)(cL + \mathcal{B}(L + x)). \end{aligned}$$

A recursion is the alternating sequence H, L, H, L, H, L, H (7 evaluations); three recursions form an outer iteration (21 evaluations, 7 differentiated), exactly as in EqR. The readout decodes $N(H)$. Noise is added after every half-step.

Variants (Table 1): $x \to L$ *only* drops x from H's force (EqR's choice); *self-read* adds the updated stream to its own force input ($L \leftarrow \ldots \mathcal{B}(L + H + x)$), which breaks invertibility; *inner loop* uses EqR's order (L^6H) with self-read; *twin* doubles L into L, L' that read each other (§2.3).

2.3 The twin coupling

Proposition 6 below shows what the alternating couplings give up: because L's force does not read L, repeated L updates with H fixed converge to an explicit function of H, one block evaluation, and EqR's inner iteration $z_L \leftarrow 0.05\, z_L + 0.95\, \mathcal{B}(z_L + z_H + x)$, which solves an implicit equation for z_L, is gone. Letting the force read the updated stream (the self-read and inner-loop arms) brings the iteration back but breaks invertibility. The twin coupling keeps both: it doubles L into two copies, and each copy's force reads the *other* copy where EqR's force reads z_L itself.

Definition 3 (Twin coupling)**.** The state is $(H, L, L') \in (\mathbb{R}^{n\times d})^3$. Let $a = e^{-\gamma}$, $b = e^{-\alpha}$, $c = e^{-\delta}$ and, for the current H,

$$G_H(z) = c\, z + \mathcal{B}(z + H + x).$$

The three half-steps are

$$\begin{aligned} T_L:&\quad L \leftarrow aL + (1-a)\, G_H(L') = aL + (1-a)\big(cL' + \mathcal{B}(L' + H + x)\big),\\ T_{L'}:&\quad L' \leftarrow aL' + (1-a)\, G_H(L) = aL' + (1-a)\big(cL + \mathcal{B}(L + H + x)\big),\\ T_H:&\quad H \leftarrow bH + (1-b)\big(cm + \mathcal{B}(m)\big), \qquad m = \tfrac{1}{2}(L + L'). \end{aligned}$$

A latent recursion is $T_L, T_{L'}, T_L, T_{L'}, T_L, T_{L'}, T_H$: EqR's order $L^6 H$, with the six L-type updates alternating between the copies (Figure 1). Fresh noise $\xi \sim \mathcal{N}(0, \sigma^2 I)$ is added to the updated stream after every half-step. An outer iteration is three recursions (21 block evaluations, the last 7 differentiated) and the readout decodes $N(H)$, as for the other couplings.

The arm with $\delta = 2$. The rates are fixed (not learned): $\gamma = \alpha = 3$ and $\delta = 2$, so $a = b = e^{-3} = 0.0498$ and $c = e^{-2} = 0.135$. Written out,

$$\begin{aligned} L &\leftarrow 0.050\, L + 0.129\, L' + 0.950\, \mathcal{B}(L' + H + x), \qquad H \leftarrow 0.050\, H + 0.129\, m + 0.950\, \mathcal{B}(m),\\ L' &\leftarrow 0.050\, L' + 0.129\, L + 0.950\, \mathcal{B}(L + H + x), \end{aligned}$$

next to EqR's $z_L \leftarrow 0.05\, z_L + 0.95\, \mathcal{B}(z_L + z_H + x)$ and $z_H \leftarrow 0.05\, z_H + 0.95\, \mathcal{B}(z_H + z_L)$. The keep-rate $a = 0.0498$ is EqR's $1 - \lambda = 0.05$ to three digits, so the twin changes EqR's update in three places only:

(i) each copy's force reads the other copy ($\mathcal{B}(L' + H + x)$ where EqR has $\mathcal{B}(z_L + z_H + x)$);

(ii) each copy also receives a cross term $(1-a)c\, L' = 0.129\, L'$, the common-mode damping of relax (Prop. 5). If the copies coincide the update is the self-reading map $L \leftarrow rL + (1-a)\mathcal{B}(L + H + x)$ with retention $r = a + (1-a)c = 0.178$; with $\delta = 1$, $r = 0.40$;

(iii) H's force reads only the copies' mean, $\mathcal{B}(m)$, not H itself (EqR's reads $z_H + z_L$), plus the cross term $0.129\, m$. As in EqR, x enters only through the L updates.

Everything else is EqR's. The block $\mathcal{B}$ is EqR's L-level module (two post-norm layers, each a token-mixing MLP and a SwiGLU with expansion 4, $d = 512$, no positional encoding) with EqR's initialisation and parameter count; the input embedding, the 16 puzzle-embedding slots, the output head and the halting head are EqR's. The coupled model keeps its states in float32 and evaluates $\mathcal{B}$ in bfloat16, and its heads read $N(H)$ (RMS normalisation without gain), the only changes common to all coupled arms. The copies are stored side by side as one $z_L \in \mathbb{R}^{2n\times d}$ (concatenated along the token axis, $z_L = [L; L']$), so EqR's carry, ACT halting (at most 16 steps in training), detached carry between outer iterations, random reset and evaluator need no change; at a reset H, L and L' are drawn independently (truncated normal, std 1). Training is EqR's recipe (Adam-atan2 with $\beta = (0.9, 0.95)$, learning rate 10^{-4} constant after 2000 warm-up steps, weight decay 1.0, batch 768, EMA 0.999, halting exploration 0.1) at $\sigma = 0.01$. Per outer iteration the twin evaluates $\mathcal{B}$ 21 times on $n = 97$ tokens, exactly as EqR; it carries $3nd$ state numbers instead of $2nd$.

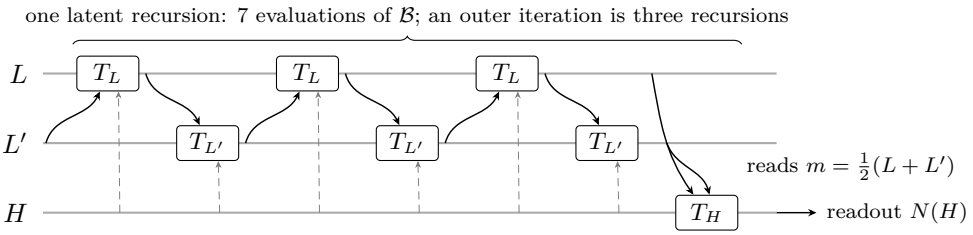


Figure 1: One latent recursion of the twin coupling. Time runs left to right; each box updates the stream on whose line it sits. Solid arrows: the copy that the update reads (its force is $\mathcal{B}(\text{copy} + H + x)$, plus the cross term $c\cdot$copy). Dashed: the current H, read by every L-type update. T_H reads only the mean of the copies. No update reads the stream it changes.

Algorithm 1 One outer iteration (ACT step) of the twin coupling

Require: carry (H, L, L'), input embedding x, noise σ; $a = e^{-\gamma}$, $b = e^{-\alpha}$, $c = e^{-\delta}$

for $r = 1, 2, 3$ **do** ▷ latent recursions; gradients flow only through $r = 3$
  for $k = 1, 2, 3$ **do**
    $L \leftarrow aL + (1-a)\big(cL' + \mathcal{B}(L' + H + x)\big) + \xi$ ▷ $\xi \sim \mathcal{N}(0, \sigma^2 I)$, fresh each time
    $L' \leftarrow aL' + (1-a)\big(cL + \mathcal{B}(L + H + x)\big) + \xi$ ▷ reads the L just computed
  $m \leftarrow \frac{1}{2}(L + L')$
  $H \leftarrow bH + (1-b)\big(cm + \mathcal{B}(m)\big) + \xi$
return cell logits and halting logit from $N(H)$; carry (H, L, L') forward, detached

2.4 Evaluation

Depth D is the number of outer iterations; *breadth* B is the number of random starts per puzzle. For $B > 1$ an answer is chosen by: the most converged start (EqR's score, the logit change over the last three iterations; "converged"); the mean accuracy of the four most converged ("top-4", EqR's headline); plurality of full grids ("vote"); the largest halting logit; or a label-free *Sudoku verifier* (a grid with no rule violations that keeps the givens; Sudoku-Extreme puzzles have unique solutions, so this equals "some start is right", "any"). Test noise defaults to $\sigma = 0.5$ as in EqR's released evaluation. Sample sizes: whole test set (423k) for in-training accuracy at 16 iterations, the first 2048 puzzles for depth×breadth, and the first 256 puzzles ×16 starts ×256 iterations for the dynamical analysis.

3 Theory

3.1 State scale

Lemma 1 (Unit-RMS force)**.** *Every token row of $\mathcal{B}(u)$ has RMS exactly one (up to the normalisation ϵ).*

Proof. $\mathcal{B}$ ends with N, which divides each row by its RMS and has no gain parameter. □

Proposition 2 (Fixed-point scale)**.** *At a noise-free fixed point, with f the (unit-RMS) force evaluated there:*

(a) EqR: $z_L^* = \mathcal{B}(\cdot)$, so $\mathrm{rms}(z_L^*) = 1$, and likewise z_H^*;

(b) kick: $L^* = \frac{a}{1-a} f_L$ and $H^* = \frac{1}{1-b} f_H$, so $\mathrm{rms}(L^*) = \frac{a}{1-a}$, $\mathrm{rms}(H^*) = \frac{1}{1-b}$;

(c) relax with merged streams $H^* = L^* = z^*$: $z^* = \frac{1}{1-c}\,\mathcal{B}(z^* + x)$, so $\mathrm{rms}(z^*) = \frac{1}{1-c}$; for $c = 1$ there is no fixed point unless $\mathcal{B}(z^* + x) = 0$.

Proof. Set new state equal to old state in each update and solve; e.g. for kick $L^* = a(L^* + f_L)$ gives $L^* = \frac{a}{1-a} f_L$; for merged relax $z^* = az^* + (1-a)(cz^* + f)$ gives $(1-a)(1-c)z^* = (1-a)f$. Lemma 1 gives the RMS. □

The identity is exact per token. The measured endpoint RMS matches it to within 1% for every EqR, kick and merged-relax checkpoint (Figure 4, Table 11; it does not apply to the unmerged $x \to L$ relax arms or to $\delta = 0$, which has no fixed point): e.g. kick (positive, 20k) L: predicted 0.64, measured 0.64; merged relax $\delta = 1$: 1.58 vs. 1.58; relax $\delta = \frac{1}{4}$: 4.52 vs. 4.51; kick (fixed): 200 vs. 199.8.

Corollary 3 (Relative noise)**.** *Isotropic noise σ acts on a stream of scale s as relative noise σ/s. At $\sigma = 0.5$ this is 0.5 for EqR, ≈ 0.3 for merged relax ($\delta = 1$), and ≈ 0.8 for the L stream of positive kick at 20k (whose L grows to 0.81 by 50k as γ falls).*

3.2 Volume

Proposition 4 (Volume bound)**.** *Each coupled half-step is a shear $(s, o) \mapsto (s + g(o), o)$ composed with a constant diagonal scaling, so for one outer iteration with n_L L-half-steps and n_H H-half-steps (per channel and token) $\log|\det J| = -nd\,(n_L\gamma + n_H\alpha)$. Hence the geometric mean of $|\mu|$ over the $2nd$ eigenvalues of J is $e^{-(n_L\gamma + n_H\alpha)/2}$, and in particular $\rho(J) \geq e^{-(n_L\gamma + n_H\alpha)/2}$ at every fixed point.*

Proof. A shear has unit Jacobian determinant; the scaling contributes a^{nd} or b^{nd} per half-step (for relax the cross term $(1-a)cH$ is part of the shear). Multiply over the 21 half-steps. The spectral radius is at least the geometric mean of the moduli. □

Remark 1. The shears are symplectic only if $\partial g/\partial o$ is symmetric, which a learned block is not, so the maps are volume-contracting at a constant rate but not conformal symplectic: there is no eigenvalue pairing $\mu \leftrightarrow e^{-\cdot}/\mu$ [16], only the determinant constraint.

The bound is 2×10^{-14} per outer iteration for relax (rate 3), 9×10^{-6} and 7×10^{-7} for positive kick (20k, 50k), 0.017 for learned kick, but 0.949 for fixed kick ($\gamma = \alpha = 0.005$). Only the last is constrained: it cannot contract, and indeed its end states have $\rho = 1.01$ and it keeps a right answer only 42% of the time even at $\sigma = 0.1$ (Table 9).

3.3 The neutral mode of undamped relaxation

Proposition 5 (Common mode)**.** *With the forces frozen, the linear part of the relax half-steps acting on (L, H) per channel is $M_L = \left(\begin{smallmatrix} a & (1-a)c \\ 0 & 1 \end{smallmatrix}\right)$, $M_H = \left(\begin{smallmatrix} 1 & 0 \\ (1-b)c & b \end{smallmatrix}\right)$. For $c = 1$ both are row-stochastic, so $(1, 1)^\top$ is a fixed vector of every product: the common mode is exactly neutral ($\mu = 1$). For $c < 1$ the row sums are $a + (1-a)c < 1$ and $b + (1-b)c < 1$, so every product has spectral radius below one. With $a = b$ and merged streams the common mode is retained at the rate $r = a + (1-a)c$ per half-step.*

Proof. Row-stochastic matrices have eigenvalue 1 with eigenvector $\mathbf{1}$; a nonnegative matrix's spectral radius is at most its maximum row sum. □

Relax with $\delta = 0$ therefore drifts along the force (endpoint RMS $\approx 4 \times 10^3$, $\rho = 1.00$ at its end states) and reaches only 13%. Accuracy rises as the retention r falls (Figure 3); EqR's own self-retention is $1 - \lambda = 0.05$.

3.4 Inner fixed-point problems

Proposition 6 (Alternating couplings have an explicit inner equilibrium)**.** *With H frozen, repeated L half-steps converge geometrically at rate a to $\Lambda(H) = cH + \mathcal{B}(H + x)$ (relax) or $\Lambda(H) = \frac{a}{1-a}\mathcal{B}(H + x)$ (kick), each a single block evaluation. In EqR the inner equilibrium solves the implicit equation $z_L = \mathcal{B}(z_L + z_H + x)$, which takes iteration.*

Proof. The force on L does not depend on L, so the L update is affine in L with slope a. □

In the language of two-timescale iterations [4], EqR's fast variable tracks an implicitly defined equilibrium $\Lambda(z_H)$ between slow updates, whereas in the alternating couplings Λ is one explicit layer, so the coupled models are effectively single-latent maps (and relax's streams merge, $\cos(H, L) = 1.00$). EqR's and TRM's ablations put the cost of a single latent at ≈ 10 points [13, 14]. The twin coupling restores an implicit inner problem:

Proposition 7 (Twins are a two-copy doubling)**.** *Let $G(z) = cz + \mathcal{B}(z + H + x)$ for frozen H. The twin updates $L \leftarrow aL + (1-a)G(L')$, $L' \leftarrow aL' + (1-a)G(L)$ with $a < 1$ have a fixed point (p, q) if and only if $p = G(q)$ and $q = G(p)$, i.e. p is a fixed point of $G \circ G$. Thus the twins represent both fixed points ($p = q$) and period-2 orbits ($p \neq q$) of the self-reading map G as fixed points.*

Proof. At a fixed point $p = ap + (1-a)G(q)$, i.e. $p = G(q)$, and symmetrically $q = G(p)$. □

Remark 2. This is the construction used to analyse synchronous Hopfield dynamics: a synchronously updated symmetric network converges to fixed points or 2-cycles, and its two-copy bipartite version, updated sequentially, has exactly these as fixed points [5, 11].

3.5 Flip bifurcation of relaxed self-maps

Proposition 8 (Period doubling)**.** *Let $F(z) = (1-\lambda)z + \lambda\tilde{\mathcal{B}}(z)$ and let μ be a real eigenvalue of $D\tilde{\mathcal{B}}$ at a fixed point z^*. Then $1 - \lambda + \lambda\mu$ is an eigenvalue of $DF(z^*)$; the direction is stable iff $1 - \frac{2}{\lambda} < \mu < 1$. As μ decreases through $1 - \frac{2}{\lambda}$ ($= -1.105$ at $\lambda = 0.95$) the eigenvalue of DF crosses -1, and generically a period-2 orbit is born (flip bifurcation). Smaller step sizes $\lambda < 2/(1-\mu)$ restore stability.*

Proof. $DF = (1-\lambda)I + \lambda D\tilde{\mathcal{B}}$ shares eigenvectors with $D\tilde{\mathcal{B}}$. The flip-bifurcation statement is the standard normal-form result for a simple eigenvalue crossing -1. □

Strong negative feedback ($\mu < -1$) is natural for Sudoku, whose constraints are mutual exclusions. The stability region and the measured values for EqR are drawn in Figure 16; damping of this kind is also the mechanism behind the fixed-point reasoners of [17].

3.6 The twin coupling as a lag-two recurrence

Proposition 9 (Invertibility, volume and scale of the twin)**.** *For the twin coupling (Definition 3):*

- *(a) each half-step is invertible, with $L = (L_{\text{new}} - (1-a)G_H(L'))/a$, $L' = (L'_{\text{new}} - (1-a)G_H(L))/a$ and $H = (H_{\text{new}} - (1-b)(cm + \mathcal{B}(m)))/b$, since T_L does not change L' or H, $T_{L'}$ does not change L or H, and T_H does not change m;*
- *(b) the Jacobian determinant is constant, $\log|\det J| = -nd\,(18\gamma + 3\alpha)$ per outer iteration on the $3nd$-dimensional state, so the geometric mean of the eigenvalue moduli is $e^{-(18\gamma+3\alpha)/3} = e^{-21} \approx 8 \times 10^{-10}$ at $\gamma = \alpha = 3$; the volume bound (Prop. 4) is not binding;*
- *(c) at a noise-free fixed point with H frozen, the copies satisfy $p = G_H(q)$, $q = G_H(p)$, and their per-token RMS lies in $[\frac{1-2c}{1-c}, \frac{1}{1-c}]$, i.e. $[0.84, 1.16]$ for $\delta = 2$ and $[0.42, 1.58]$ for $\delta = 1$; if the copies coincide it equals $\frac{1}{1-c}$ exactly (1.16 for $\delta = 2$), as for merged relax (Prop. 2c).*

Proof. (a) Each update is affine in the stream it changes, with slope a or b, and its other terms are functions of unchanged streams. (b) Each $T_L, T_{L'}$ scales nd coordinates by a and each T_H scales nd by b; an outer iteration has 18 of the former and 3 of the latter. (c) The fixed-point equations follow as in Proposition 7. Per token, $p = cq + f$ with $\mathrm{rms}(f) = 1$ (Lemma 1), so $|\,\mathrm{rms}(p) - 1| \le c\,\mathrm{rms}(q)$ and symmetrically; hence $\mathrm{rms}(p) \le 1 + c + c^2\,\mathrm{rms}(p)$, giving $\mathrm{rms}(p) \le \frac{1}{1-c}$, and then $\mathrm{rms}(p) \ge 1 - \frac{c}{1-c}$. For $p = q$, $(1-c)p = f$. □

Proposition 10 (Lag-two recurrence)**.** *Fix H and write the copies' values as one interleaved sequence: before a recursion $(L, L') = (z_{-1}, z_0)$, and the j-th L-type update writes z_j (into L for odd j, into L' for even j). Then the six L-type updates of a recursion are*

$$z_{j+1} = a\, z_{j-1} + (1-a)\, G_H(z_j), \qquad j = 0, \dots, 5,$$

and T_H reads $m = \frac{1}{2}(z_5 + z_6)$, the mean of the last two iterates. In particular:

(a) for $a = 0$ the inner loop is plain iteration $z_{j+1} = G_H(z_j)$ of the self-reading map, i.e. EqR's inner loop with $\lambda = 1$ and a cross term cz; the twin's $a = 0.0498$ relaxes towards the iterate two *steps back instead of the previous one;*

(b) at a fixed point $z^* = G_H(z^*)$ (coinciding copies), along an eigenvector of $DG_H(z^*)$ with real eigenvalue μ the per-update multipliers s solve $s^2 - (1-a)\mu\, s - a = 0$. Both roots lie in $(-1, 1)$ if and only if $-1 < \mu < 1$, and their product is $-a$, the volume constraint of Proposition 9(b);

(c) at $\mu = -1$ the roots are -1 and a. The critical mode alternates in sign along the sequence, so it is antisymmetric in the copies ($\delta L = -\delta L'$), and over one $(T_L, T_{L'})$ sweep its multiplier is $s^2 = 1$. The period doubling of the self-map (Prop. 8) therefore appears in the twin state as a symmetry-breaking pitchfork: the copies separate, $(L, L') \to (p, q)$ with $p \ne q$ a 2-cycle of G_H (Prop. 7), and H reads the stationary mean $\frac{1}{2}(p + q)$.

Proof. The recurrence is T_L and $T_{L'}$ rewritten in the interleaved indices. (b) Linearising, $e_{j+1} = (1-a)\mu e_j + a e_{j-1}$. With $f(s) = s^2 - (1-a)\mu s - a$, $f(0) = -a < 0$, $f(1) = (1-a)(1-\mu)$ and $f(-1) = (1-a)(1+\mu)$, so there is one root in each of $(-1, 0)$ and $(0, 1)$ iff $-1 < \mu < 1$; the product of the roots is $f(0) = -a$. (c) $f(s) = (s+1)(s-a)$ at $\mu = -1$; the eigen-solution $e_j \propto (-1)^j$ puts opposite signs on consecutive iterates, i.e. on L and L'. □

The step $(z_{j-1}, z_j) \mapsto (z_j, z_{j+1})$ is the additive coupling of reversible residual networks [12] with a scaling a, and the same device that makes momentum ResNets invertible [18]: a first-order self-map that cannot be inverted is made invertible by carrying two consecutive iterates. The merged fixed point of the twin loses stability at $\mu(DG_H) = -1$, i.e. at $\mu(D\mathcal{B}) = -1 - c = -1.14$ for $\delta = 2$, close to EqR's flip threshold $1 - \frac{2}{\lambda} = -1.105$; but what forms then is a stationary pair rather than an oscillation, and H sees both phases through m. Whether trained twins sit on the diagonal $L = L'$ or off it has not been measured yet (the dynamical analysis of §6 covers rounds one and two only).

3.7 Certification and noise

EqR bounds the distance to a fixed point by the residual under local contraction ($\|z - z^*\| \le \|R(z)\|/(1 - L)$) and turns this into a correctness certificate when the output margin is large [13, App. A.1]. The certificate needs right answers to be contracting and wrong ones not to be; we measure exactly this in §6.2. For noise, along a normal eigen-direction with eigenvalue μ the linearised stationary variance is $\sigma^2/(1 - |\mu|^2)$: near-critical directions amplify noise, and a basin is left once the fluctuations reach its boundary. In Hopfield networks spurious mixture states lose stability at a lower temperature than retrieval states [1, 6]; noise then selects the right attractors. Section 6.3 tests this.

Table 2: Accuracy at training noise ($\sigma = 0.01$), one start. Exact@16: whole test set; @64/@256: first 49,152 puzzles. O: right at 16 but wrong at 256; Δy: prediction changed between 16 and 256 (not computed for round three, last block).

arm	steps	ex@16	cell@16	ex@64	ex@256	$O_{16\to256}$	$\Delta y_{16\to256}$
EqR	20k	69.5	89.3	73.2	75.8	1.7	31.3
EqR	50k	85.8	94.7	92.1	95.0	0.4	14.8
relax, $\delta = 1$	20k	63.9	87.3	65.4	66.4	2.2	36.6
relax, $\delta = 1$	50k	78.3	91.7	81.8	84.5	1.3	23.5
kick, positive rates	20k	63.5	87.2	65.0	66.1	1.9	35.8
kick, positive rates	50k	78.2	91.7	81.7	83.9	1.4	23.2
relax, $\delta = \frac{1}{4}$	20k	46.7	82.3	47.8	48.6	3.8	32.7
kick, learned rates	20k	46.5	82.1	47.3	48.0	2.0	37.2
relax, $\delta = 0$	20k	13.2	71.0	12.8	3.5	80.3	97.3
kick, fixed rates	20k	13.5	72.6	14.7	6.2	61.3	94.6
relax, $\delta = 1$, $x \to L$ only	20k	51.7	84.0	52.2	52.7	2.7	38.6
relax, $\delta = 1$, $x \to L$, self-read	20k	61.7	86.7	63.5	64.9	1.7	35.9
kick, positive, $x \to L$ only	20k	56.2	85.1	56.7	57.8	1.9	42.0
relax, $\delta = 2$	20k	65.8	88.1	68.0	69.9	–	–
relax, $\delta = 2$	50k	70.3	88.2	72.4	74.6	–	–
relax, $\delta = 3$	20k	63.5	87.4	65.2	66.7	–	–
twin L, $\delta = 1$	20k	62.8	87.1	65.6	67.5	–	–
twin L, $\delta = 2$	20k	61.4	86.8	63.6	65.2	–	–
twin L, $\delta = 2$	50k	84.6	94.3	90.7	93.7	–	–
inner loop, self-read, $\delta = 1$	20k	65.2	87.8	72.0	76.0	–	–
inner loop, self-read, $\delta = 1$	50k	81.0	93.1	90.0	94.0	–	–

4 Accuracy and training

Table 2 and Figure 2 give the accuracies. Our EqR reproduces the paper (85.8 at 16 iterations and 92.1 at 64 after 50k steps, against 86.4 and 93.0 [13]). At 20k steps the two best couplings, relax $\delta = 1$ and positive kick, trail EqR by about six points at 16 iterations; the gap does not close with training (7.5 points at 50k) and widens with depth (10.5 points at 256 iterations). Fixed small rates fail (Prop. 4, 5).

Round two: input injection and self-reading. Giving x only to L (as EqR does) costs 12 points for relax and 7 for kick. For relax it separates the streams ($\cos(H, L)$ 1.00 $\to$ 0.77) but each stream is then driven only by the other. Adding self-reading recovers 10 of the 12 points (51.7 $\to$ 61.7), but only to the level of plain merged relax (63.9), which already reads itself in effect (its streams coincide). Self-reading without an inner loop is therefore not what EqR has over relax.

Round three: damping and inner loops. Tables 2, 4 and 3, Figures 2 and 3. At 20k and the training noise, lowering the merged-state retention from 0.40 ($\delta = 1$) to 0.18 ($\delta = 2$) gives the best coupled arm at 16 iterations (65.8, against 63.9), and $\delta = 3$ ($r = 0.10$) is worse again. The twins keep H apart from L ($\cos = 0.25$–0.45) but are not better than relax at 20k (62.8, 61.4), and only the inner-loop arm gains clearly with depth (65.2 $\to$ 76.0 from 16 to 256 iterations; EqR 69.5 $\to$ 75.8). Three arms were trained on to 50k, and there the order changes (Figure 2a): the twin with $\delta = 2$ rises from 62.3 at 21k to 85.0 at 49k, against EqR's 85.9, and the inner loop reaches 80.6, while relax $\delta = 2$ peaks at 74.5 at 35k and falls to 71.1. At 50k the twin's depth curve at the training noise is within 1.5 points of EqR's (84.6/90.7/93.7 against 85.8/92.1/95.0 at 16/64/256 iterations), the closest any coupling has come.

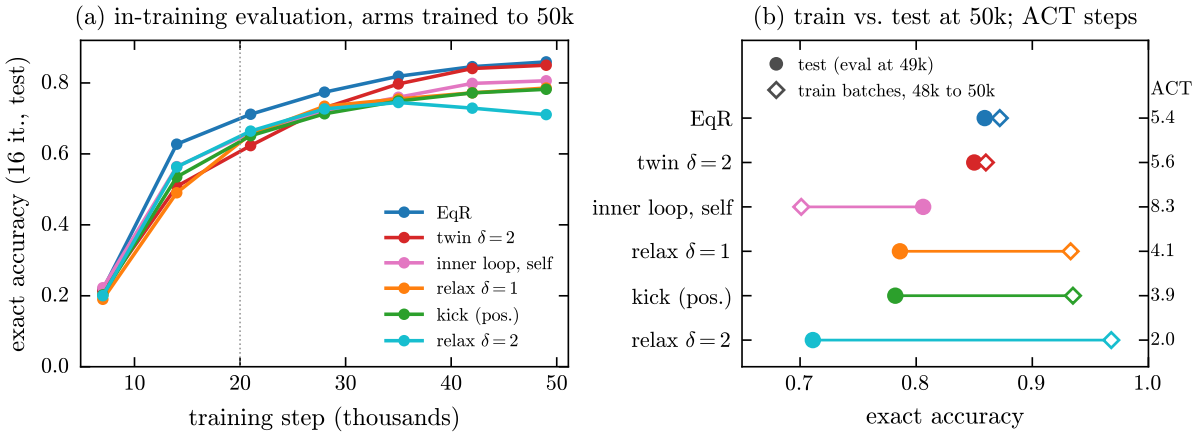


Figure 2: (a) In-training evaluation (whole test set, 16 iterations, $\sigma = 0.01$) of the six arms trained to 50k steps. The twin ($\delta = 2$) starts slowly and nearly reaches EqR; relax $\delta = 2$ peaks at 35k. (b) Test accuracy at 49k (filled) against training-batch accuracy averaged over steps 48–50k (open), with the mean number of ACT steps on training batches at the right.

Table 3: Generalisation at 50k (W&B). Test: in-training evaluation (whole test set, 16 iterations, $\sigma = 0.01$) at 49k and its maximum over the evaluations every 7k steps. Train: exact accuracy of training batches averaged over steps 48–50k; it counts sequences that halt, so it depends on the halting head. ACT: mean outer iterations per training sequence. $\cos(H, L)$ over the same window.

arm	test 49k	best test	train 48–50k	train−test	ACT	$\cos(H, L)$
EqR	85.9	85.9	87.2	+1.3	5.4	0.50
relax, $\delta = 1$	78.6	78.6	93.3	+14.7	4.1	0.99
kick, positive rates	78.2	78.2	93.5	+15.3	3.9	0.75
relax, $\delta = 2$	71.1	74.5 (35k)	96.8	+25.7	2.0	0.98
twin L, $\delta = 2$	85.0	85.0	86.0	+1.0	5.6	0.41
inner loop, self-read, $\delta = 1$	80.6	80.6	70.1	−10.5	8.3	0.71

Generalisation. Table 3 and Figure 2b. The training set of Sudoku-Extreme-1k is 1000 puzzles (each with 1000 augmentations), so a model can fit it. At 48–50k the alternating couplings are right on 93–97% of training batches but on 71–79% of test puzzles, and relax $\delta = 2$ halts after 2.0 outer iterations on training puzzles, the fewest of all arms, while its test accuracy falls. EqR and the twin show gaps of about one point (87.2 against 85.9; 86.0 against 85.0) and halt after 5.4–5.6 iterations. The self-reading inner loop is right less often on training batches than on the test set (70.1 against 80.6, with 8.3 ACT steps), which says more about its halting head than about fitting. At 20k all gaps were small (EqR +1, relax $\delta = 1$ +3 points), so the gap opens with training. The arms whose inner equilibrium is a single explicit layer (Prop. 6) thus fit the training puzzles, and the three arms with an implicit inner problem (EqR, twin, inner loop) do not. This is a correlation across six single-seed runs, not a controlled test.

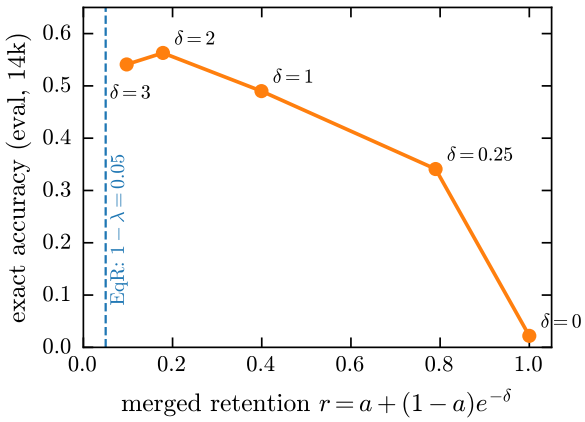


Figure 3: Relax, rate 3: accuracy at 14k against the retention r of the merged state (Prop. 5).

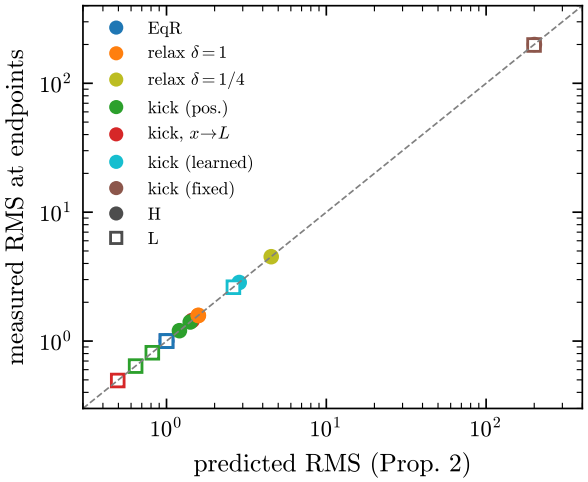


Figure 4: Endpoint state RMS against Prop. 2 (H filled, L open); all trained arms with a fixed point.

Table 4: Round three at 20k (W&B). Eval: whole test set, 16 iterations, at 14k. Train: training-batch exact accuracy averaged over 18–20k (depends on halting). $\cos(H, L)$ and mean ACT steps over the same window.

arm	eval@14k	train 18–20k	$\cos(H, L)$	ACT steps
EqR	62.7	72.0	0.41	7.9
relax, $\delta = 1$	49.0	68.4	1.00	8.2
kick, positive rates	53.5	64.8	0.58	8.8
relax, $\delta = \frac{1}{4}$	34.1	44.9	1.00	11.1
relax, $\delta = 2$	56.3	74.2	1.00	7.3
relax, $\delta = 3$	54.1	70.6	1.00	7.9
twin L, $\delta = 1$	50.4	62.4	0.45	9.3
twin L, $\delta = 2$	50.8	65.1	0.25	8.9
inner loop, self-read, $\delta = 1$	56.4	59.0	0.72	9.9

5 Test-time scaling

Figure 5 and Table 5 (upper block). All working models improve steadily with depth. At 20k, relax ties EqR once compute is large (depth 1024: 90.4 vs. 91.4; depth 64 with 128 starts, most converged: 95.9 vs. 95.7). At 50k EqR leads everywhere. Relax's error decays like $D^{-0.39}$ without saturating, so it would need more than an order of magnitude more iterations to match EqR at 50k. Round three, and the twin in particular, is in §5.1.

Positive kick at 20k is the outlier (Figure 6): at $\sigma = 0.5$ a single run is right only 37% of the time, yet some start is right for 99.1% of puzzles and a plurality vote over 128 starts reaches 96.8%, the best of the round-one 20k models. Its relative noise is about 1.6 times EqR's (Corollary 1), which pushes it into a high-diversity regime; at noise 0.25 its single run is 69%. By 50k its L stream has grown (Prop. 2) and the regime is gone (any 93.8, vote 87.9).

5.1 Round three: the twin coupling scales best at 20k

Figure 7 and the lower block of Table 5. At 20k the twin with $\delta = 2$ has the worst single run at 16 iterations of all working arms (52.3%) and the steepest depth curve we have measured: error 47.7% → 20.4% → 7.4% → 1.8% at depths 16, 64, 256 and 1024, a power law with exponent −0.78 (EqR at 20k: −0.31; the released EqR: −0.56; relax $\delta = 1$: −0.35). At depth 1024 it is right on 98.2% of puzzles, against 91.4% for EqR at 20k and 97.6% for EqR's released 50k

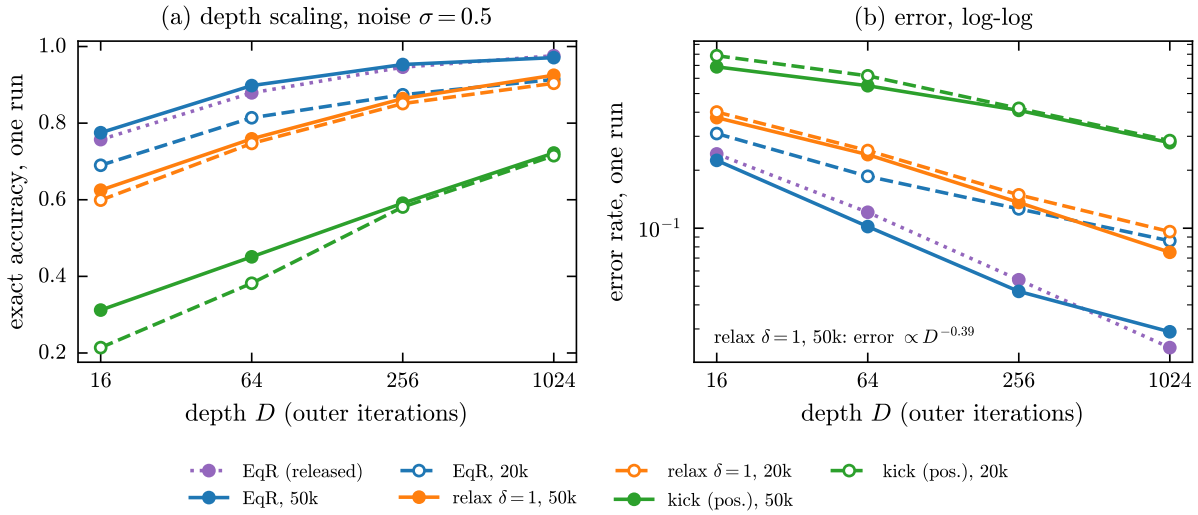


Figure 5: One-run accuracy against depth at $\sigma = 0.5$ (first 2048 test puzzles). Dashed/open: 20k steps; solid/filled: 50k; dotted: EqR's released checkpoint. (b) The error of relax falls like a power of depth; EqR's falls faster and then flattens near 3%.

checkpoint. In EqR's headline setting, depth 64 with 128 starts, it reaches 99.6% top-4, 99.6% for the most converged start and 99.8% by plurality vote (released EqR: 98.6, 99.2, 98.7); some start is right for 99.9% of puzzles, and convergence finds it. On 2048 puzzles that is about 4 wrong by vote against 27. The twin with $\delta = 1$ is in between: 91.6% at depth 1024, level with EqR at 20k, and 95.5% top-4. So for the twins stronger common-mode damping pays at depth. For relax under test noise it does not: at depth 1024, $\delta = 1, 2, 3$ give 90.4, 88.3 and 78.4%, with depth exponents -0.35, -0.30 and -0.20, although $\delta = 2$ was the best relax arm at the training noise.

The gain comes from test noise. Table 6 and Figure 8. At the training noise the 20k twin is a weak model (65.2% at depth 256, below relax $\delta = 2$); at $\sigma = 0.5$ it is right on 92.6% at the same depth, a gain of 27 points, the largest we have measured. Twin $\delta = 1$ and relax $\delta = 1$ gain about 18 points, EqR 12. The noise does not help the self-reading inner loop at all (76.0 against 75.8), so that arm's strong depth scaling at the training noise (§4) does not carry over to EqR's protocol: at $\sigma = 0.5$ it is the second-weakest round-three arm (81.8% at depth 1024). Positive kick loses from noise at every checkpoint. By the escape analysis (§6.3), noise below an arm's breakdown level $\sigma^\star$ destabilises wrong end states without losing right ones; a gain as large as the twin's means that at 20k many of its wrong end states are shallow while its right ones survive $\sigma = 0.5$. We have not yet run the escape analysis on twin checkpoints, so this is an inference from the accuracies.

Longer training removes the need for noise. At 50k every arm is accurate at the training noise, and $\sigma = 0.5$ helps EqR by 0.3 points and hurts all others: the twin by 3.1, the inner loop by 3.7, kick by 25, and relax $\delta = 2$ collapses (2.6% exact at every depth), the collapse that relax $\delta = 1$ showed at $\sigma = 0.75$ (Figure 9) now at a lower noise. The 50k twin is better than the 20k twin at short depth (68.0 against 52.3% at 16 iterations) but worse at large depth (95.4 against 98.2% at 1024; most converged of 16 starts at depth 64: 96.0 against 98.3%), and EqR at 50k leads it at every depth. A fixed σ is a different test for each checkpoint (Corollary 1 and §6.3); whether the 50k twin recovers its depth scaling at a smaller test noise is the first thing to measure next.

Table 5: Test-time scaling at $\sigma = 0.5$, first 2048 test puzzles. Left: one start at depth D. Right: depth 64 with 128 starts, by selection rule (best non-oracle rule in bold; "any" is the oracle). The released EqR gives 98.6 top-4, as in EqR's README. Lower block: round three; "–": depth 64 with 128 starts at 50k still running.

		one run, depth D				depth 64, 128 starts			
arm	steps	16	64	256	1024	converged	top-4	vote	any
EqR	20k	69.0	81.4	87.4	91.4	**95.7**	94.0	94.3	95.9
EqR	50k	77.5	89.8	95.3	97.1	**98.7**	98.0	98.0	98.8
EqR (released)	50k	75.7	87.9	94.6	97.6	**99.2**	98.6	98.7	99.2
relax, $\delta = 1$	20k	59.9	74.7	85.1	90.4	**95.9**	93.4	93.8	96.0
relax, $\delta = 1$	50k	62.5	75.9	86.4	92.5	**96.6**	95.3	95.6	96.7
kick, positive rates	20k	21.4	38.2	58.1	71.5	86.3	85.1	**96.8**	99.1
kick, positive rates	50k	31.2	45.1	59.1	72.2	**93.7**	88.5	87.9	93.8
relax, $\delta = 2$	20k	59.3	72.5	82.2	88.3	**95.0**	92.4	92.4	95.0
relax, $\delta = 2$	50k	2.6	2.6	2.6	2.7	–	–	–	–
relax, $\delta = 3$	20k	49.4	62.3	71.2	78.4	**87.9**	84.0	84.4	88.0
twin L, $\delta = 1$	20k	58.3	75.8	85.9	91.6	**97.3**	95.5	96.3	97.5
twin L, $\delta = 2$	20k	52.3	79.6	92.6	98.2	99.6	99.6	**99.8**	99.9
twin L, $\delta = 2$	50k	68.0	81.1	90.6	95.4	–	–	–	–
inner loop, self-read, $\delta = 1$	20k	54.2	67.7	75.8	81.8	**88.7**	85.6	86.6	89.4
inner loop, self-read, $\delta = 1$	50k	70.0	83.8	90.3	93.3	–	–	–	–

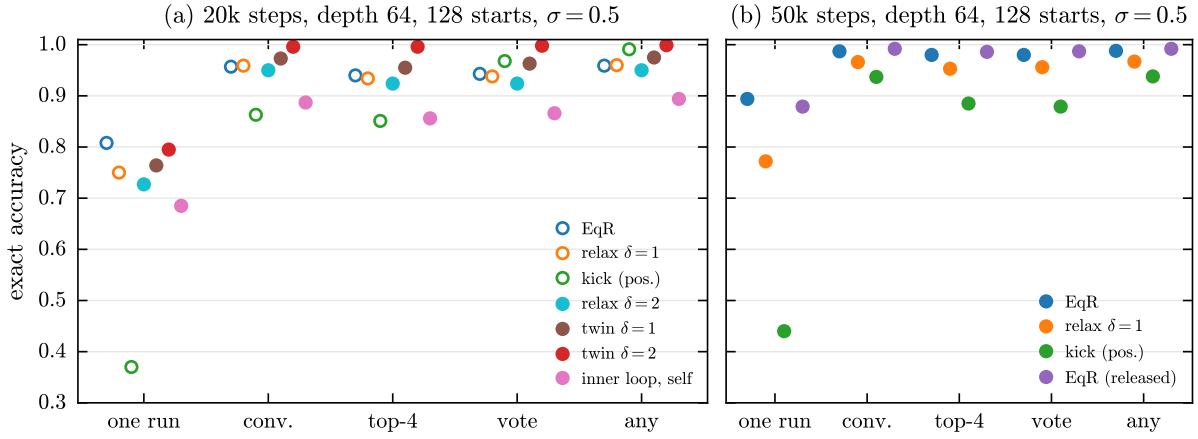


Figure 6: Depth 64, 128 starts, $\sigma = 0.5$: accuracy by selection rule (conv.: most converged start; top-4: mean of the four most converged; any: some start right). (a) 20k, round-one arms open, round-three arms filled: the twin with $\delta = 2$ is best under every selection rule (EqR has the better single run). Among the round-one arms positive kick has the best vote and "any" but the worst single run and convergence selection; (b) by 50k this has reversed. The round-three arms at 50k are still being evaluated.

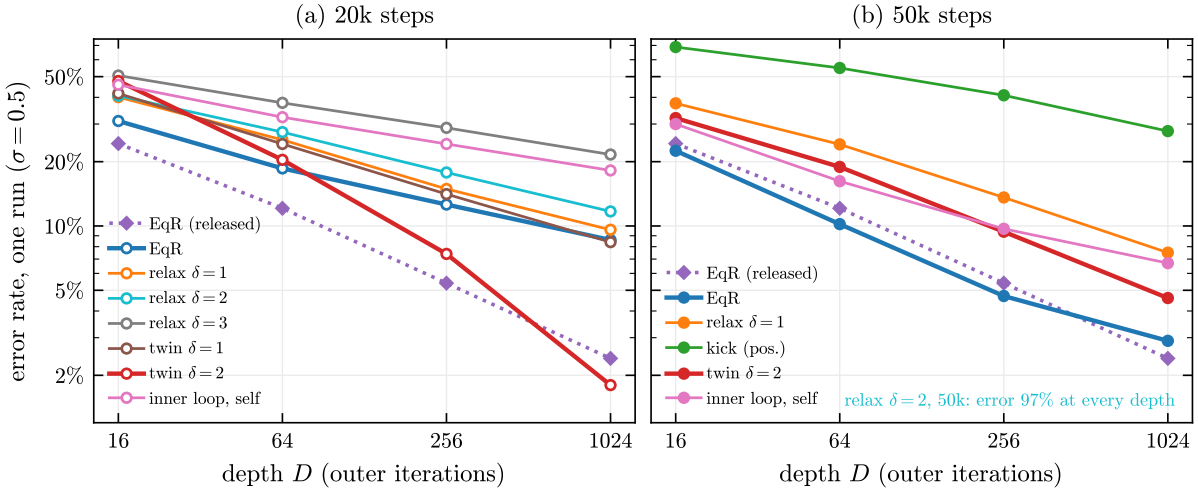


Figure 7: Round three: one-run error against depth at $\sigma = 0.5$ (first 2048 test puzzles), log–log. Dotted: EqR's released 50k checkpoint in both panels. (a) At 20k the twin with $\delta = 2$ starts worst and ends best: its error falls 26-fold from depth 16 to 1024 ($D^{-0.78}$), against 3.6-fold for EqR at 20k and 10-fold for the released checkpoint. (b) At 50k the twin is faster at short depth but its curve flattens ($D^{-0.47}$); EqR at 50k leads at every depth.

Table 6: Exact accuracy (%) of one run at the training noise $\sigma = 0.01$ (Table 2; depth 16 on the whole test set, depth 256 on 49,152 puzzles) and at the test noise $\sigma = 0.5$ (Table 5; first 2048 puzzles). Δ: effect of the test noise at depth 256. The puzzle sets differ, so differences below about one point are not meaningful.

		depth 16		depth 256			depth 1024
arm	steps	$\sigma = 0.01$	0.5	$\sigma = 0.01$	0.5	Δ	$\sigma = 0.5$
EqR	20k	69.5	69.0	75.8	87.4	+11.6	91.4
EqR	50k	85.8	77.5	95.0	95.3	+0.3	97.1
relax, $\delta = 1$	20k	63.9	59.9	66.4	85.1	+18.7	90.4
relax, $\delta = 1$	50k	78.3	62.5	84.5	86.4	+1.9	92.5
kick, positive rates	20k	63.5	21.4	66.1	58.1	−8.0	71.5
kick, positive rates	50k	78.2	31.2	83.9	59.1	−24.8	72.2
relax, $\delta = 2$	20k	65.8	59.3	69.9	82.2	+12.3	88.3
relax, $\delta = 2$	50k	70.3	2.6	74.6	2.6	−72.0	2.7
relax, $\delta = 3$	20k	63.5	49.4	66.7	71.2	+4.5	78.4
twin L, $\delta = 1$	20k	62.8	58.3	67.5	85.9	+18.4	91.6
twin L, $\delta = 2$	20k	61.4	52.3	65.2	92.6	+27.4	98.2
twin L, $\delta = 2$	50k	84.6	68.0	93.7	90.6	−3.1	95.4
inner loop, self-read, $\delta = 1$	20k	65.2	54.2	76.0	75.8	−0.2	81.8
inner loop, self-read, $\delta = 1$	50k	81.0	70.0	94.0	90.3	−3.7	93.3

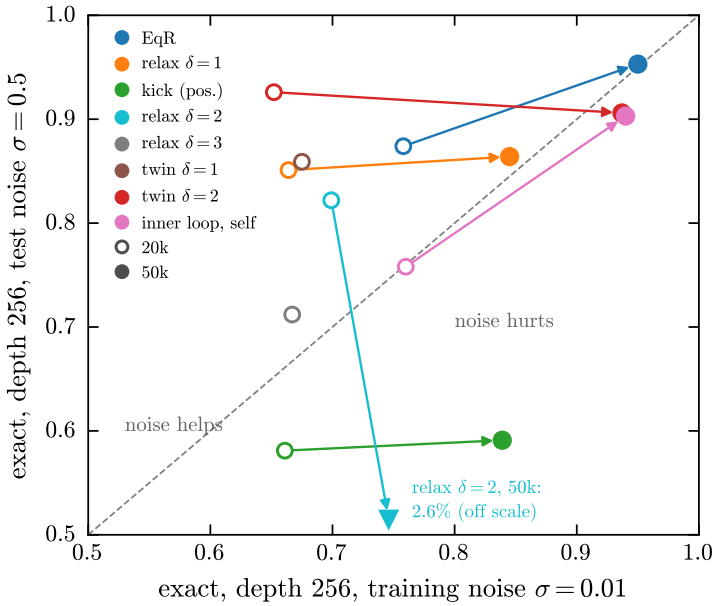


Figure 8: Depth-256 accuracy at the training noise against the test noise $\sigma = 0.5$; open: 20k, filled: 50k, arrows join the two checkpoints of one arm. Above the diagonal noise helps. At 20k the twins and relax $\delta = 1$ gain most; by 50k every arm is near or below the diagonal, and relax $\delta = 2$ collapses (off scale).

6 Dynamics of the trained maps

All quantities in this section use the noise-free map unless stated, in float32, on the first 256 test puzzles with 16 random starts and 256 outer iterations, at the checkpoints' EMA weights. The analysis covers the round-one and round-two checkpoints; the round-three checkpoints (twin, inner loop, relax $\delta \in \{2, 3\}$) have not been analysed yet.

6.1 Selection is solved; coverage is not

Table 7. For EqR and relax at $\sigma \leq 0.5$, the most converged start, the start with the largest halting logit and the verifier all agree with "any": whenever some start is right, each rule finds it, and the convergence score separates right from wrong runs with AUROC 0.98–1.00 (Figure 13). The exceptions are informative: relax $\delta = \frac{1}{4}$ at $\sigma = 0$ (AUROC 0.53; its right end states are not contracting, $\rho = 1.16$) and positive kick at 20k under $\sigma = 0.5$ (AUROC 0.86), the regime of its vote advantage. The differences between models are in *coverage* and in single-run accuracy, and these track path independence: at 50k EqR's starts reach 1.6 distinct answers per puzzle, relax's 2.6 and kick's 3.2 (Figure 9c).

6.2 Spectra at the end states

Observation 1 (Right answers attract, wrong end states do not)**.** In every working model the median spectral radius at right end states is below one and at wrong end states above one (Table 8, Figure 10). Wrong end states after 256 iterations are therefore not stable fixed points but slow transients or unstable sets, which is why convergence certifies correctness (§6.1) and why per-instance path independence separates right from wrong answers [2].

EqR's right answers sit close to criticality and move closer with training ($0.87 \to 0.93 \to 0.97$), as reported for converged Sudoku loopers [15] and in line with the drift of equilibrium models toward critical stability [3]. The coupled maps contract much harder (0.53–0.73). Leading eigenvalues at right answers are complex for EqR and relax at 20k (88%) but real at 50k (13%, 0%), and real for positive kick throughout (Figure 11): kick is *not* oscillatory at its answers, contrary to a momentum (underdamped) picture [10].

One caveat qualifies the comparison. For the coupled maps the spectral radius is almost the same at every right end state ($q_{10} \approx q_{90}$ in Table 8, e.g. $[0.53, 0.53]$ for relax at 20k), whereas EqR's varies from puzzle to puzzle. A puzzle-independent mode of the coupled update, rather than the puzzle's own constraint dynamics, may therefore set the coupled maps' ρ; the leading Ritz values (Figure 11) are real and tightly clustered, consistent with this. Identifying that mode is left open.

6.3 Escape under noise

Table 9 and Figure 12. Up to an arm-specific level $\sigma^\star$ right answers are kept with probability one while 15–33% of wrong end states are rescued; rescue peaks just below $\sigma^\star$. This is the selective destabilisation of spurious attractors known from Hopfield networks [1, 6], and it matches the published observation that noise lowers single-path quality but raises best-of-N [9]. $\sigma^\star$ is above 1 for EqR at 50k and relax at 20k, between 0.75 and 1 for EqR at 20k, the released EqR and kick at 20k, and between 0.5 and 0.75 for relax and kick at 50k. The best test noise for breadth is just below $\sigma^\star$, so it must be chosen per checkpoint: relax tolerated $\sigma = 0.75$ at 20k and collapses there at 50k (single run 3.7%), although its state scale is unchanged; its right answers attract less strongly at 50k (ρ 0.53 $\to$ 0.68). Relax $\delta = \frac{1}{4}$ rescues almost nothing: its wrong attractors are deep, which is why breadth does not help it.

Table 7: Multistart (256 puzzles, 16 starts, depth 256). "conv.": most converged; "q": largest halting logit; verifier: valid Sudoku keeping the givens. AUROC of EqR's convergence score for right vs. wrong runs. "answers": mean number of distinct final grids per puzzle.

arm	steps	single	any	vote	conv.	q	verifier	AUROC	answers	ΔPI
test noise $\sigma = 0$										
EqR	20k	73.5	83.2	79.3	83.2	83.2	83.2	0.984	4.5	0.038
EqR	50k	96.3	98.8	98.4	98.8	98.8	98.8	1.000	1.6	0.027
EqR (released)	50k	96.9	99.6	99.2	99.6	99.6	99.6	1.000	1.5	0.020
relax, $\delta = 1$	20k	64.1	70.7	68.8	70.7	70.7	70.7	1.000	5.2	0.043
relax, $\delta = 1$	50k	89.7	94.1	93.4	94.1	94.1	94.1	0.999	2.6	0.033
kick, positive rates	20k	63.4	70.3	66.8	70.3	70.3	70.3	0.998	4.0	0.036
kick, positive rates	50k	85.4	92.2	90.2	92.2	92.2	92.2	1.000	3.2	0.040
relax, $\delta = \frac{1}{4}$	20k	48.4	52.0	49.2	50.8	52.0	52.0	0.532	2.3	0.020
test noise $\sigma = 0.5$										
EqR	20k	89.4	97.3	96.9	97.3	97.3	97.3	0.999	2.7	0.050
EqR	50k	96.2	99.2	98.8	99.2	99.2	99.2	0.995	1.6	0.030
EqR (released)	50k	95.9	99.6	99.6	99.6	99.6	99.6	1.000	1.6	0.034
relax, $\delta = 1$	20k	82.6	94.1	91.4	94.1	94.1	94.1	0.997	3.7	0.075
relax, $\delta = 1$	50k	88.1	99.2	98.4	98.4	99.2	99.2	0.995	2.9	0.083
kick, positive rates	20k	63.0	97.7	91.8	85.2	71.5	97.7	0.864	6.8	0.236
kick, positive rates	50k	64.1	89.5	82.8	89.1	89.5	89.5	0.997	6.6	0.140
relax, $\delta = \frac{1}{4}$	20k	56.7	65.6	62.1	65.6	64.8	65.6	0.996	7.6	0.034

6.4 Inside one outer iteration

Figure 14. EqR at 20k shows HRM's *hierarchical convergence* [19]: within each recursion the L change falls roughly by half per update, then H moves and L restarts. The coupled maps have no such structure (Prop. 6); every update changes the state by a similar amount, and the second of two consecutive H updates at each recursion boundary changes almost nothing, so 2 of the 21 evaluations per outer iteration are nearly wasted by the alternating schedule.

6.5 EqR's inner loop goes through a flip bifurcation

Figures 15 and 16. At 20k the inner fixed point is stable (leading eigenvalues of DF near ± 0.6 for right puzzles; wrong puzzles sit at -1.00 to -1.03). At 50k a cluster of real eigenvalues sits at -0.93 to -1.07, and in the released checkpoint the leading eigenvalue is ≈ -1.9 to -2.0, i.e. $\mu \approx -2.1$ for the block. The inner fixed point is then unstable and z_L lives on a stable 2-cycle (two-step Jacobian eigenvalues 0.91–0.99). Because six inner updates are an even number, z_H always reads the same phase; in the released model z_H is then stationary to 4×10^{-7}.

The cycle carries no information the readout uses: odd inner-loop lengths, which make z_H read alternating phases, are as accurate as even ones (Table 10; ± 2.5 points at 128 puzzles). We read the period doubling as a by-product of strong negative feedback in the block combined with EqR's light damping, not as a computational mechanism. By Propositions 7 and 10 a twin coupling holds such a cycle as a fixed point with $L \neq L'$; the cosine between the copies is therefore a direct readout of whether the twin arms learn the same structure, and it is the first quantity to measure on the twin checkpoints.

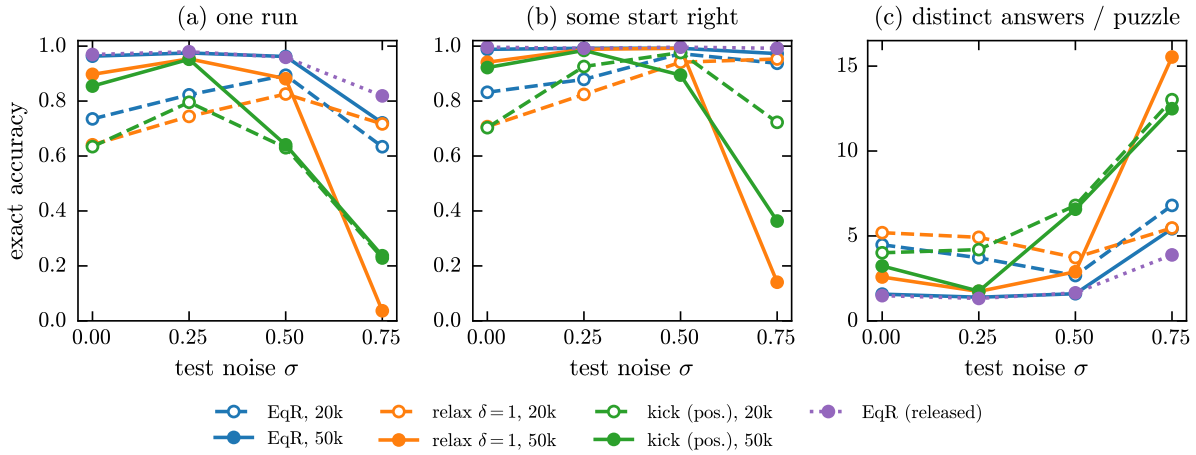


Figure 9: Multistart against test noise (256 puzzles, 16 starts, depth 256). (a) one run; (b) some start right; (c) distinct answers per puzzle. Relax at 50k collapses at $\sigma = 0.75$ although it tolerated it at 20k.

Table 8: Spectral radius of one outer iteration's Jacobian at noise-free end states (16 right and 16 wrong per arm, power iteration on JVPs), and the share of complex leading Ritz values (Arnoldi, 30 steps).

		right end states				wrong end states	
arm	steps	ρ	$[q_{10}, q_{90}]$	$\mathbb{C}$ top-1	$\mathbb{C}$ top-10	ρ	$\mathbb{C}$ top-1
EqR	20k	0.87	[0.81, 0.87]	88	85	1.52	25
EqR	50k	0.93	[0.90, 1.60]	12	71	4.76	44
EqR (released)	50k	0.97	[0.96, 0.99]	6	39	7.96	31
relax, $\delta = 1$	20k	0.53	[0.53, 0.53]	88	79	1.63	50
relax, $\delta = 1$	50k	0.67	[0.67, 0.67]	0	59	2.59	56
relax, $\delta = 1$, $x \to L$, self-read	20k	0.39	[0.39, 0.39]	0	40	2.41	38
relax, $\delta = \frac{1}{4}$	20k	1.16	[0.82, 2.35]	44	61	1.98	38
relax, $\delta = 0$	20k	1.00	[1.00, 1.00]	0	51	1.00	6
kick, positive rates	20k	0.73	[0.73, 0.74]	0	32	3.32	62
kick, positive rates	50k	0.66	[0.66, 0.66]	0	9	2.36	31
kick, learned rates	20k	1.26	[1.04, 1.72]	94	76	2.38	38
kick, fixed rates	20k	1.01	[0.99, 1.04]	0	46	0.99	0

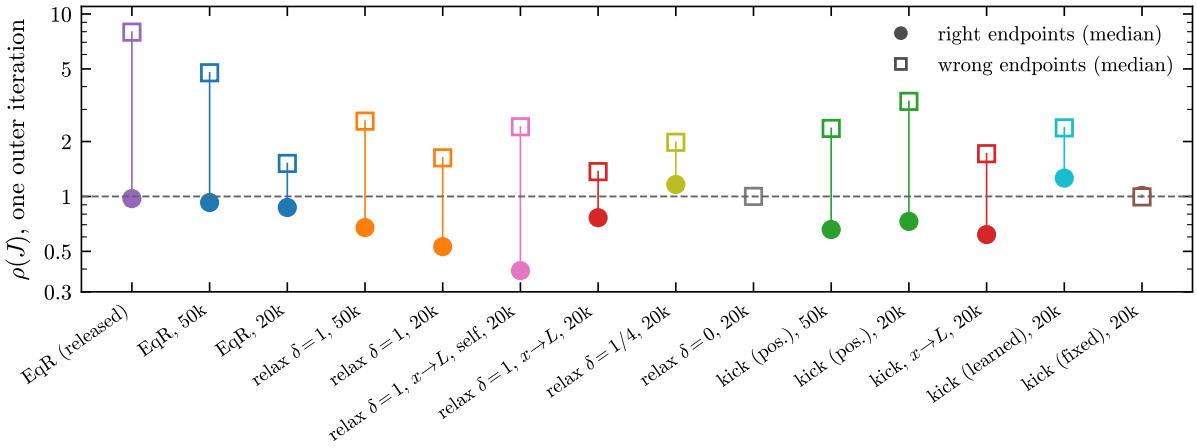


Figure 10: Median spectral radius of one outer iteration at right (filled) and wrong (open) end states. Dashed: $\rho = 1$.

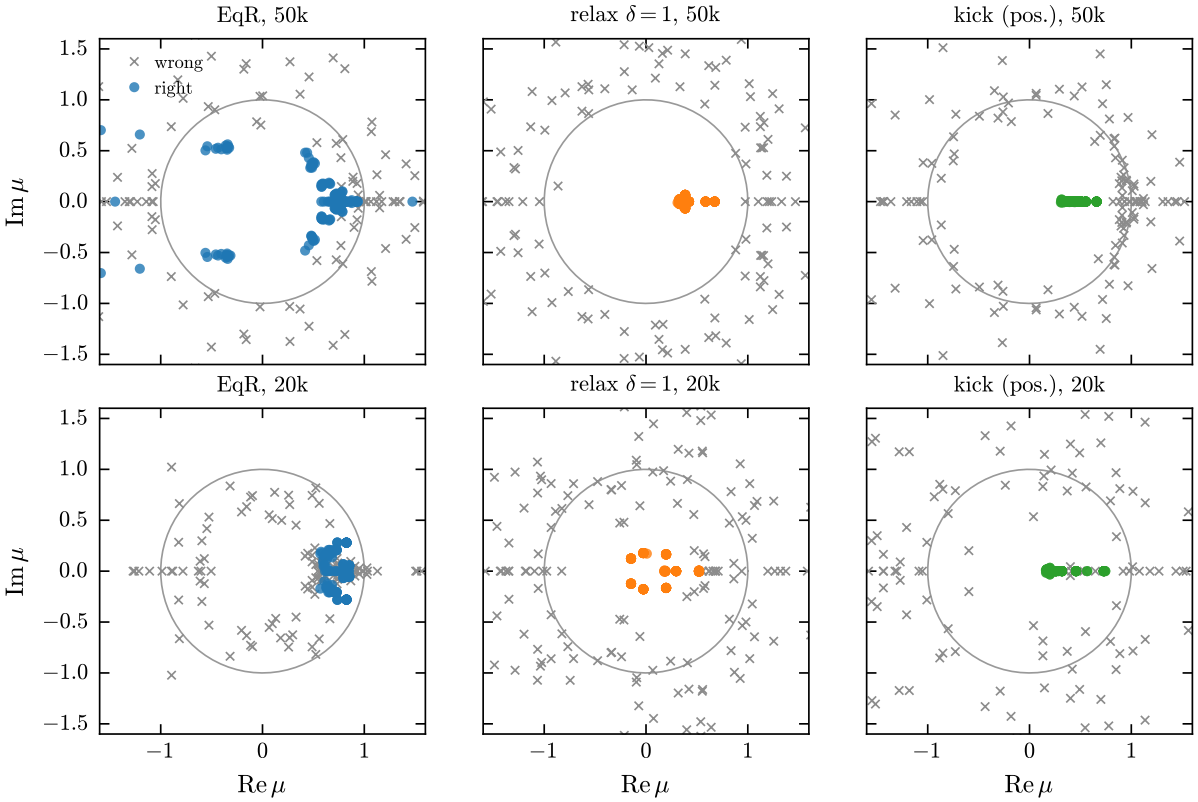


Figure 11: Leading Ritz values (top 10 per end state) of one outer iteration's Jacobian. Coloured: right end states; grey crosses: wrong end states; circle: $|\mu| = 1$.

Table 9: From noise-free end states, $K = 32$ iterations at noise σ: $P(\text{keep} \mid \text{right})$ / $P(\text{right} \mid \text{wrong})$.

arm	steps	$\sigma = 0.1$	0.25	0.5	0.75	1.0
EqR	20k	1.00 / 0.03	1.00 / 0.13	1.00 / 0.30	0.92 / 0.15	0.04 / 0.00
EqR	50k	1.00 / 0.07	1.00 / 0.16	1.00 / 0.27	1.00 / 0.04	0.98 / 0.03
EqR (released)	50k	1.00 / 0.06	1.00 / 0.14	1.00 / 0.23	0.99 / 0.11	0.08 / 0.00
relax, $\delta = 1$	20k	1.00 / 0.05	1.00 / 0.12	1.00 / 0.23	1.00 / 0.14	1.00 / 0.05
relax, $\delta = 1$	50k	1.00 / 0.07	1.00 / 0.24	1.00 / 0.15	0.04 / 0.00	0.02 / 0.00
kick, positive rates	20k	1.00 / 0.08	1.00 / 0.23	1.00 / 0.17	1.00 / 0.02	0.03 / 0.00
kick, positive rates	50k	1.00 / 0.07	1.00 / 0.33	1.00 / 0.17	0.33 / 0.00	0.01 / 0.00
relax, $\delta = \frac{1}{4}$	20k	1.00 / 0.02	1.00 / 0.06	1.00 / 0.07	1.00 / 0.05	1.00 / 0.04
relax, $\delta = 0$	20k	0.75 / 0.00	0.76 / 0.00	0.76 / 0.00	0.74 / 0.00	0.75 / 0.00
kick, fixed rates	20k	0.41 / 0.00	0.42 / 0.00	0.42 / 0.00	0.42 / 0.00	0.42 / 0.00

Table 10: Inner-loop length at test time (128 puzzles, one start, 32 outer iterations, local float32): exact accuracy at $\sigma = 0$ / $\sigma = 0.5$. Six is the trained value.

checkpoint	L=4	5	6	7	8
EqR 20k	75.0 / 81.2	75.0 / 80.5	71.1 / 79.7	73.4 / 79.7	72.7 / 78.1
EqR 50k	86.7 / 87.5	91.4 / 89.1	92.2 / 88.3	90.6 / 86.7	90.6 / 86.7
EqR released	90.6 / 89.1	94.5 / 85.9	94.5 / 84.4	93.0 / 88.3	94.5 / 85.2

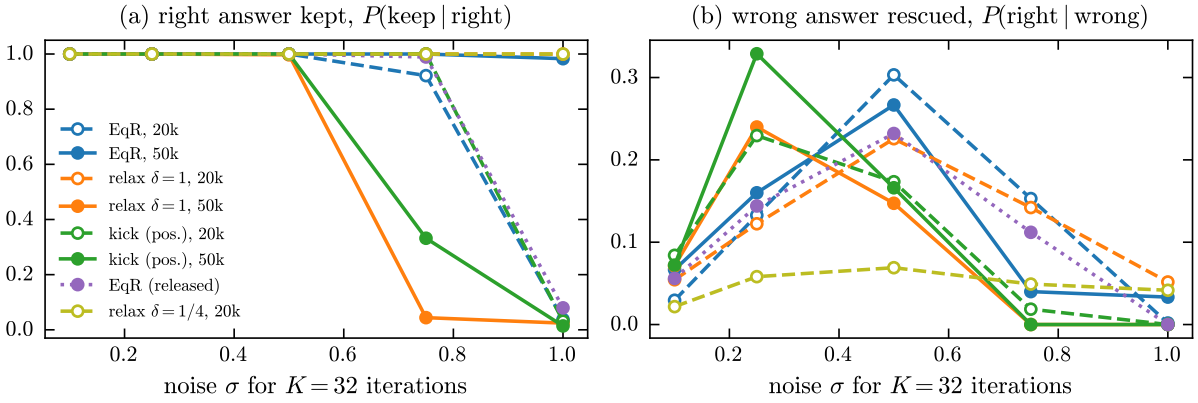


Figure 12: Escape from noise-free end states after $K = 32$ noisy iterations. (a) right answers kept; (b) wrong answers rescued.

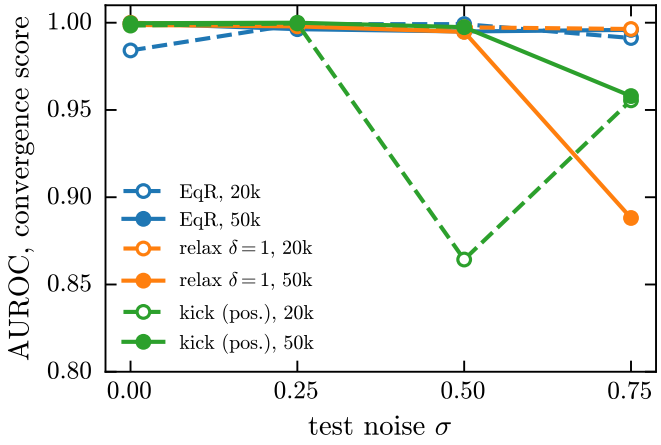


Figure 13: AUROC of EqR's convergence score for right vs. wrong runs, against test noise.

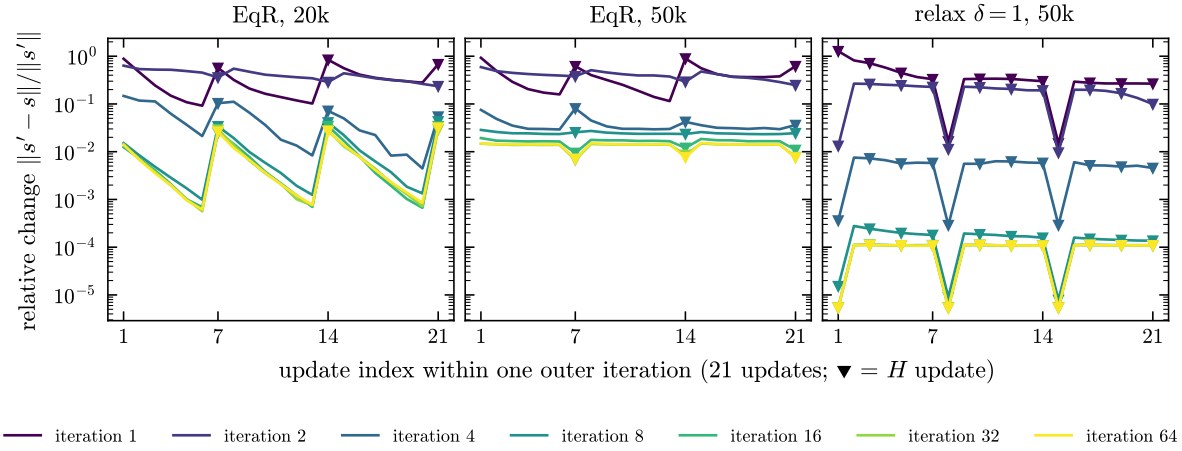


Figure 14: Median relative change per update inside one outer iteration (21 updates, ▼ = H updates), at outer iterations $1, \ldots, 64$ from a random start (64 puzzles). EqR at 20k shows hierarchical convergence: L settles within each recursion and is reset by H. At 50k the L changes no longer decay (the 2-cycle of §6.5). For relax the dips are the second of two consecutive H updates at recursion boundaries; the floor near 10^{-4} is TF32 round-off on the GPU.

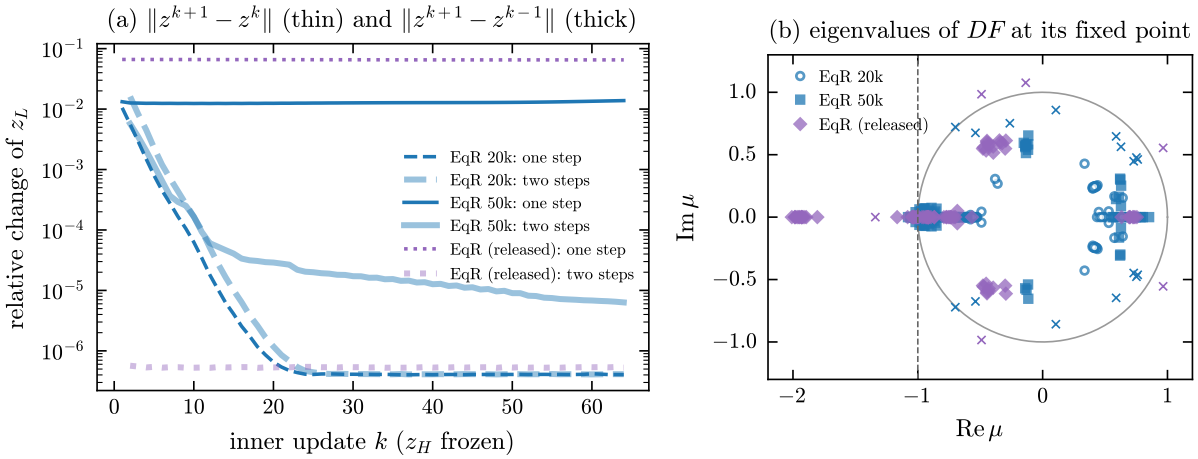


Figure 15: EqR's inner map $F(z_L) = 0.05 z_L + 0.95\mathcal{B}(z_L + z_H + x)$ with z_H frozen at an end state (16 puzzles, local float32). (a) Median one-step (thin) and two-step (thick) relative change over 64 inner updates: at 20k both vanish (fixed point); at 50k and in the released model the one-step change stays at 1% and 6% while the two-step change drops to round-off (exact 2-cycle). (b) Leading eigenvalues of DF at its fixed point (found by damped iteration); crosses: wrong puzzles.

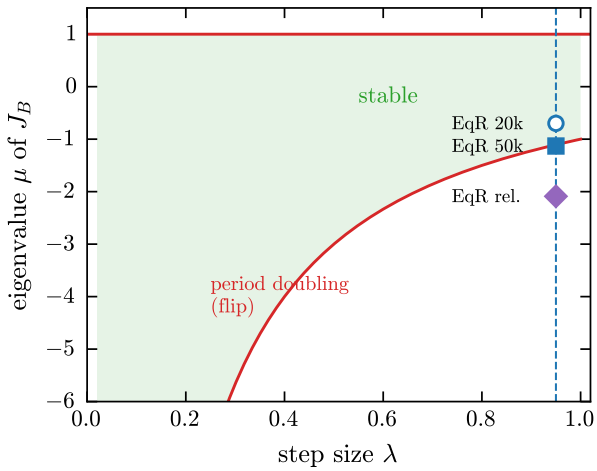


Figure 16: Stability of $F = (1-\lambda)z + \lambda\tilde{\mathcal{B}}(z)$ along a real eigen-direction μ of $D\tilde{\mathcal{B}}$ (Prop. 8). Markers: most negative leading eigenvalue of DF at the inner fixed point (median over right puzzles), mapped to μ at EqR's $\lambda = 0.95$.

7 Discussion

Which properties matter.

1. *Attracting right answers and non-attracting wrong ones.* Every working model has it, and it makes selection by convergence or halting essentially free. Models without it (relax $\delta = 0$, $\frac{1}{4}$; fixed and learned kick) also lose the certificate.
2. *Large, dominant basins for right answers (path independence).* This is where EqR leads, and it grows with training for every arm.
3. *A margin between the noise that breaks wrong attractors and the noise that breaks right ones.* It exists for all working models; its location $\sigma^\star$ is set by the state scale (exactly predictable, Prop. 2) and by contraction strength, and it moves with training.
4. *Damping of the common mode*, not the volume constraint. The undamped mode of relax is fatal; strong but not maximal damping is best.
5. *An implicit inner equilibrium problem.* EqR has one; the alternating couplings, whose inner equilibrium is a single explicit layer, fit the 1000 training puzzles and stop improving on the test set (train−test gaps of 15–26 points at 50k). Restoring the inner problem without self-reading, by the twin's two copies, removes the gap (one point at 50k) and brings the twin to within 1–1.5 points of EqR at the training noise. This suggests that invertibility is not what costs the alternating couplings accuracy, and that the missing inner loop is.
6. *Noise-driven exploration that pays off with depth*, which depends on the checkpoint. At 20k the twin gains 27 points from test noise at depth 256 and beats EqR's released 50k checkpoint at depth 1024 and with 128 starts; the self-reading inner loop, which also has an implicit inner problem, gains nothing. At 50k noise $\sigma = 0.5$ helps only EqR. What makes a map profit from noise is not yet measured; the twin's lag-two structure (Prop. 10) and the escape and spectral analysis of its checkpoints are where to look.

Limitations. All runs use one seed; in particular the twin's 99.6% at depth 64 with 128 starts has not been replicated. Test-time numbers use 2048 puzzles (±0.5 points; ±0.2 near 99%) and the dynamical analysis 256 puzzles (±1–2 points). Spectra use 16 right and 16 wrong end states per arm, and Ritz values from 30 Arnoldi steps approximate only the leading spectrum. GPU traces ran with TF32 matrix multiplies (round-off $\approx 10^{-4}$); the inner-loop analysis was redone locally in float32. The round-three checkpoints have test-time results but no dynamical analysis, and their depth-64, 128-start evaluation at 50k is still running. All test-time comparisons use EqR's $\sigma = 0.5$, which is not the best noise for every checkpoint. Training-batch accuracy depends on halting, so the generalisation gaps are indicative.

Next. (i) Seeds 1 and 2 of the twin ($\delta = 2$) and EqR at 20k, with depth 1024 and depth 64×128 starts. (ii) A test-noise sweep at 50k ($\sigma \in \{0.1, 0.25, 0.75\}$) for the twin, the inner loop, relax $\delta = 2$ and EqR. (iii) The dynamical analysis of the twin checkpoints at 20k and 50k: whether the copies coincide, spectra at right and wrong end states, and escape under noise. (iv) Twins with larger δ or learned rates, and a schedule without consecutive H updates for the alternating couplings.

A Rates, scales and dimensionality

Table 11: Rates, the volume bound per outer iteration (Prop. 4), predicted vs. measured endpoint RMS of H / L (Prop. 2), and participation ratios $(\sum \ell_i)^2 / \sum \ell_i^2$ of the per-token state covariance.

arm	steps	γ	α	δ	vol. bound	RMS pred.	RMS meas.	PR H / L
EqR	20k	$1-\lambda = 0.05$ (self-read)			–	1.00 / 1.00	1.00 / 1.00	11.2 / 14.2
EqR	50k	$1-\lambda = 0.05$ (self-read)			–	1.00 / 1.00	1.00 / 1.00	10.5 / 14.4
EqR (released)	50k	$1-\lambda = 0.05$ (self-read)			–	1.00 / 1.00	1.00 / 1.00	10.1 / 12.5
relax, $\delta = 1$	20k	3.00	3.00	1.00	2.1×10^{-14}	1.58 / 1.58	1.58 / 1.58	14.3 / 14.3
relax, $\delta = 1$	50k	3.00	3.00	1.00	2.1×10^{-14}	1.58 / 1.58	1.58 / 1.58	12.8 / 12.8
relax, $\delta = \frac{1}{4}$	20k	3.00	3.00	0.25	2.1×10^{-14}	4.52 / 4.52	4.51 / 4.51	14.1 / 14.1
relax, $\delta = 0$	20k	3.00	3.00	–	2.1×10^{-14}	∞ / ∞	4028 / 4027	12.2 / 12.2
kick, positive rates	20k	0.94	1.24	–	8.5×10^{-6}	1.41 / 0.64	1.41 / 0.64	14.1 / 13.3
kick, positive rates	50k	0.80	1.77	–	6.5×10^{-7}	1.20 / 0.82	1.20 / 0.81	13.3 / 14.3
kick, learned rates	20k	0.32	0.43	–	1.7×10^{-2}	2.85 / 2.62	2.85 / 2.62	14.3 / 17.5
kick, fixed rates	20k	0.005	0.005	–	9.5×10^{-1}	201 / 200	200 / 198	28.5 / 36.9
kick, positive, $x \to L$ only	20k	1.11	1.18	–	6.0×10^{-6}	1.45 / 0.50	1.44 / 0.49	14.0 / 13.6

B Reproduction

- Training and evaluation: `jobs/looped/sudoku_eqr.slurm`, `jobs/looped/sudoku_scaling.slurm`; summary tables: `looped/eqr/summarize_sudoku.py`.
- Dynamical analysis: `looped/eqr/export_for_analysis.py`, `looped/eqr/analyze_maps.py` (GPU job `jobs/looped/sudoku_maps.slurm`), `looped/eqr/summarize_maps.py`.
- This note: `looped/eqr/paper/make_figures.py` and `make_tables.py`, from the data in `looped/eqr/paper/data/`.

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